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CENTRIFUGAL HEAT DISSIPATION FAN

Abstract

Provided is a centrifugal heat dissipation fan including a housing and an impeller. The impeller is disposed in the housing. The impeller has a hub and multiple blades disposed surrounding the hub. Every two adjacent blades have different air guiding inclined edges such that the air guiding inclined edges pass by a fixed position of the housing and generate blade tones of varying frequencies when the impeller rotates. Each of the air guiding inclined edges approaches the housing along with a radial direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation application of and claims the priority benefit of a prior application Ser. No. 18/302,783, filed on Apr. 18, 2023, now allowed, which claims the priority benefit of Taiwan application Serial No. 111114837, filed on Apr. 19, 2022. The entirety of each of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

[0002] The disclosure relates to a heat dissipation fan, and more particularly to a centrifugal heat dissipation fan.

Description of Related Art

[0003] With the development of technology, portable electronic devices such as notebook computers and smart phones have been widely used in daily life. While electronic devices are improved to meet people's requirements for small size and high performance, how to deal with the thermal energy generated during operation is a major issue for improving the operational performance of electronic devices. Therefore, a heat dissipation module or a heat dissipation element such as a heat dissipation fan is usually disposed in an electronic device to help dissipate the thermal energy generated during operation out of the electronic device.

[0004] With a centrifugal impeller as an example, in the method of manufacturing a centrifugal impeller, a hub and fan blades are usually integrally formed through plastic injection, metal die casting, or sheet metal stamping. If the fan blades are plastic, it is difficult to reduce the thickness of the fan blades due to the material properties, which leads to a smaller spacing between two fan blades and thus reduces the heat dissipation performance of the impeller. As a result, the only way to enhance the air pressure and air volume of the centrifugal impeller is to increase the rotational speed. However, the increased rotational speed of the impeller in turn generates more noise.

[0005] Therefore, how to improve the impeller's related structure to improve its heat dissipation performance and reduce the noise it generates is actually an issue that people skilled in the art have to consider and solve.

SUMMARY

[0006] The disclosure provides a centrifugal heat dissipation fan, with blades thereof having different blade structures and thus generating blade tones of varying frequencies during rotation to reduce noise.

[0007] The centrifugal heat dissipation fan of the disclosure includes a housing and an impeller. The impeller is disposed in the housing. The impeller has a hub and multiple blades disposed surrounding the hub. Every two adjacent blades have different blade structures relative to the housing such that the blade structures pass by a fixed position of the housing and generate blade tones of varying frequencies when the impeller rotates.

[0008] Based on the above, with the blades of the centrifugal heat dissipation fan having different

blade structures, which pass by the fixed position of the housing and generate the blade tones of varying frequencies when the impeller rotates at a fixed speed, noise energy at fixed frequencies or multiples thereof can be avoided to be accumulated such that the desired effect of fan blade noise reduction can be achieved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is an exploded view of a centrifugal heat dissipation fan according to an embodiment of the disclosure.

[0010] FIG. 2 is a partial front view of an impeller in FIG. 1.

[0011] FIG. 3 is a schematic view of the design basis of a blade structure.

[0012] FIG. 4A is an exploded view of a centrifugal heat dissipation fan according to another embodiment of the disclosure.

[0013] FIG. 4B is a partial enlarged view of an impeller in FIG. 4A.

[0014] FIG. 4C is a schematic view of a blade of the impeller in FIG. 4A.

[0015] FIG. 4D is a top view of the blade in FIG. 4C.

DESCRIPTION OF THE EMBODIMENTS

[0016] FIG. 1 is an exploded view of a centrifugal heat dissipation fan according to an embodiment of the disclosure. FIG. 2 is a partial front view of an impeller in FIG. 1. With reference to FIG. 1 and FIG. 2 together, in this embodiment, a centrifugal heat dissipation fan **100** includes a housing **110** and an impeller **120**. The housing **110** includes a top plate **111** and a base plate **112**. The top plate **111** is disposed with an air inlet **E1**, and the base plate **112** is disposed with an air inlet **E2**. After the top plate **111** and the base plate **112** are combined, an air outlet **E3** is formed. The impeller **120** is disposed in the housing **110** and located between the top plate **111** and the base plate **112**. The impeller **120** has a hub **121** and multiple blades **122** disposed surrounding the hub **121**. The impeller **120** has a rotation axis **L1** in the housing **110**, so that when the impeller **120** rotates, the air in the external environment is sucked into the housing **110** through the air inlets **E1** and **E2**, and is exhausted from the housing **110** through the air outlet **E3** to perform heat dissipation on a target object, such as a processor, a display chip, or a heat dissipation fin disposed in an electronic device.

[0017] As mentioned above, when an impeller in the prior art rotates at a fixed rotational speed, blades sweep through the housing space and generate blade tones of a fixed frequency, causing energy accumulation and generating noise. Therefore, in order to effectively reduce the blade noise, in the impeller **120** of this embodiment, every two adjacent blades **122** have different blade structures relative to the housing **110**. In this way, when the impeller **120** rotates, these blade structures pass through a fixed position of the housing **110** and generate blade tones of varying frequencies to avoid tone energy accumulation at the same frequency.

[0018] With reference to FIG. 2 again, only one single blade **122** is shown for illustration. In this embodiment, the aforementioned blade structure is the blade edge of the blade **122**, which includes an upper blade edge **122a** and a lower blade edge **122b** of the blade **122** as shown in the figure. Furthermore, each blade **122** has a combining section **S1** and an air suction section **S2**. The combining section **S1** is combined with the hub **121**. For example, by in-mold injection combining the hub **121** with the combining section **S1** when the hub **121** is formed, the combining section **S1** is embedded in the hub **121**, and the structure of the hub **121** can pass through an opening **122e** of the blade **122** to enhance the combining strength of the two. Moreover, the air suction section **S2** extends from the combining section **S1** in a direction along a radial direction **R1** relative to the rotation axis **L1**. During extension, the size of the air suction section **S2** gradually increases along the rotation axis **L1**. In other words, the size of the combining section **S1** along the rotation axis **L1**

is smaller than the size of the air suction section S2 along the rotation axis L1, so that the air suction section S2 has air guiding inclined edges 122c and 122d near the combining section S1. Herein, the air guiding inclined edge 122c is located on the upper blade edge 122a of the blade 122, and the air guiding inclined edge 122d is located on the lower blade edge 122b of the blade 122. [0019] Accordingly, the purpose of making every two adjacent blades 122 have different blade structures mentioned above is to allow a designer to provide the air guiding inclined edges 122c and 122d in different sizes for different blades 122. As shown in FIG. 2, the blades 122 are provided with the air guiding inclined edges 122c and 122d in different sizes, which means the blade edges of the blades 122 keep different spacings from the housing 110. With FIG. 2 as an example, for the upper blade edge 122a, changing the size of the air guiding inclined edge 122c is to change a spacing d1 between the upper blade edge 122a and the top plate 111, while changing the air guiding inclined edge 122d is to change a spacing d2 between the lower blade edge 122b and the base plate 112.

[0020] In detail, since the air guiding inclined edges 122c and 122d respectively correspond to the air inlets E1 and E2, tapered spaces SP1 and SP2 are formed by the blade structures (the air guiding inclined edges 122c and 122d), the top plate 111, and the base plate 112 of the housing 110. The tapered spaces SP1 and SP2 taper from the air inlets E1 and E2 of the centrifugal heat dissipation fan 100 relative to the rotation axis L1 of the impeller 120 in the radial direction R1. In this way, when the impeller 120 rotates about the rotation axis L1, the blades 122 sweep through the space in the housing 110 and generate blade tones in the tapered spaces SP1 and SP2. In other words, the air guiding inclined edges 122c and 122d pass through the fixed position (i.e., the inner walls of the tapered spaces SP1 and SP2 to which the top plate 111 and the base plate 112 respectively correspond near the air inlets E1 and E2) of the housing 110 and generate blade tones.

[0021] Herein, the air guiding inclined edges 122c and 122d are inclined relative to the rotation axis L1 and respectively form included angles $\theta_{\text{sub}.n}$ and $\alpha_{\text{sub}.n}$. Therefore, changing the included angles $\theta_{\text{sub}.n}$ and $\alpha_{\text{sub}.n}$ can accordingly adjust the areas of the blades 122 in regions A1 and A2, and can accordingly change the spacings d1 and d2 of the blades 122 relative to the housing 110 as described above.

[0022] Based on the above, once the two adjacent blades 122 have different structural features at the air guiding inclined edges 122c and 122d, the frequency of the blade tones can be changed accordingly to avoid generating the blade tones of a fixed frequency and prevent energy accumulation of the blade tones. In this way, it is equivalent to accumulating the original blade tones at a fixed frequency into noise with greater energy and breaking it into blade tones with less energy due to frequency variation, which may naturally achieve the noise reduction effect for human hearing.

[0023] FIG. 3 is a schematic view of the design basis of a blade structure. With reference to FIG. 2 and FIG. 3 together, for further illustration, although generating the blade tones of varying frequencies as mentioned above can have noise reduction effect, the manufacturing process and efficiency of the blades 122 also need to be considered. Accordingly, the sizes of the blade structure in this embodiment substantially vary periodically with in accordance with the sequence in which the blades 122 surround the hub 121. With the sine wave function period shown in FIG. 3 as an example, in the design process, the included angles $\theta_{\text{sub}.n}$ and $\alpha_{\text{sub}.n}$ of the air guiding inclined edges 122c and 122d relative to the rotation axis L1 vary periodically according to the positions of the blades 122. In this embodiment, FIG. 3 only takes the included angle $\theta_{\text{sub}.n}$ as an example. As shown in FIG. 1, 59 blades 122 are disposed surrounding the hub 121, and the included angles $\theta_{\text{sub}.n}$ of these blades 122 at the air guiding inclined edge 122c may form a sequence $(\theta_{\text{sub}.n}) = (\theta_{\text{sub}.1}, \theta_{\text{sub}.2}, \theta_{\text{sub}.3}, \dots, \theta_{\text{sub}.59})$ along an annular direction (clockwise or counterclockwise). After that, the angles in the included angle sequence $(\theta_{\text{sub}.n})$ are changed in accordance with the sine wave period, and sizes d3, d4, d5 of the blades 122 are changed accordingly, with $d3 = d4 + a * |\sin \theta_{\text{sub}.n}|$ and $d5 = b * |\sin \theta_{\text{sub}.n}|$, where a and b are adjustable design

parameters. Furthermore, the number of blades **122** in this embodiment is a prime number to avoid the symmetrical effect generated by periodic frequencies, which means in this embodiment, $|\sin \theta_{\text{sub}.1}| \neq |\sin \theta_{\text{sub}.59}|$ to avoid adjacent blades **122** having the same blade structure. Accordingly, for the fixed position of the housing **110** (for example, the inner wall of the tapered space **SP1** to which the top plate **111** corresponds), each blade **122** generates blade tones of different frequencies when passing by the fixed position. Of course, this embodiment does not limit the frequency, functional form, and waveform of the periodic variation. Herein, the included angle sequence ($\theta_{\text{sub}.n}$) actually divides the 360-degree round of the impeller **120** into equal or nearly equal divisions for every two adjacent blades **122** among the blades **122** of the impeller **120** to have the same or nearly the same included angle. In addition, the gravity center of the blades **122** can be closer to the center to achieve a better balance and a better manufacturing yield. In another embodiment not shown, the included angles $\theta_{\text{sub}.n}$ of the blades may also be unequally distributed, but in that case, it is necessary to provide counterweight to the blades (such as putty or reinforcement structures on the blades) to avoid the unbalanced center of gravity. The blades here also avoid generating the noise at a fixed frequency because of the unequal distribution.

[0024] In addition, with reference to FIG. 2 again, each blade **122** in this embodiment further has an extension sub-section **S21** between the air guiding inclined edge **122c** (or **122d**) and the combining section **S1**. The extension sub-section **S21** corresponds to the air inlet **E1** or **E2**, and the size of the extension sub-section **S21** along the rotation axis **L1** is substantially equal to the size of the combining section **S1** along the rotation axis **L1**, which provides a larger space for the air inlets **E1** and **E2** to facilitate air suction from the external environment.

[0025] FIG. 4A is an exploded view of a centrifugal heat dissipation fan according to another embodiment of the disclosure. FIG. 4B is a partial enlarged view of an impeller in FIG. 4A, with part of the blade **222** removed for easy identification. With reference to FIG. 4A and FIG. 4B together, a centrifugal heat dissipation fan **200** includes a housing **210** and the impeller **220**. The housing **210** includes a top plate **211** and a base plate **212**. The top plate **211** is disposed with an air inlet **E4** along the axial direction (i.e., the rotation axis), and the base plate **212** is disposed with an air inlet **E5** along the axial direction (i.e., the rotation axis). After the top plate **211** and the base plate **212** are combined, a lateral (radial) air outlet **E6** is formed. The impeller **220** is disposed in the housing **210** and located between the top plate **211** and the base plate **212**. The impeller **220** has a hub **221** and multiple blades **222** disposed surrounding the hub **221**. As the centrifugal heat dissipation fan **100** mentioned above, in this embodiment, when the impeller **220** rotates, the air in the external environment is sucked into the housing **210** through the air inlets **E4** and **E5**, and is exhausted from the housing **210** through the air outlet **E6** to perform heat dissipation on a target object, such as a processor, a display chip, or a heat dissipation fin disposed in an electronic device.

[0026] FIG. 4C is a schematic view of a blade of the impeller in FIG. 4A. FIG. 4D is a top view of the blade in FIG. 4C. With reference to FIG. 4C and FIG. 4D together, only one single blade **222** is shown as an example, and the rest are the same as those in the previous embodiment. Here, unlike the previous embodiment where the blade edges of the blades **222** are provided as different blade structures, in this embodiment, a rib **222c** is disposed on a blade surface **222b** of the blade **222** (the rib **222c** is concave on the back of the blade surface **222b** since the rib **222c** is formed by sheet metal stamping). The ribs **222c** keep a protruding distance $t1$ from the blade surface **222b**, and then the protruding distance $t1$ of these ribs **222c** varies periodically in accordance with the sequence in which the blades **222** are disposed on the hub **221**. Accordingly, when the impeller **220** rotates, these blades **222** may also generate blade tones of varying frequencies at the fixed position of the housing **210** to achieve the desired noise reduction effect. In addition, air guiding inclined edges **222a** and **222d** of this embodiment are the same as in the previous embodiment, and details thereof are not repeated.

[0027] In summary, in the above embodiments of the disclosure, with the blades of the centrifugal heat dissipation fan having different blade structures, which pass by the fixed position of the

housing and generate the blade tones of varying frequencies when the impeller rotates at a fixed speed, noise energy at fixed frequencies or multiples thereof can be avoided to be accumulated such that the desired effect of fan blade noise reduction can be achieved.

[0028] Furthermore, in order to avoid the accumulation of noise energy at fixed frequencies or multiples thereof and also improve the manufacturing process and efficiency of the impeller, the blade structures in the disclosure are designed to show periodic variation, which means the blade structures are changed in accordance with the sequence in which the blades are disposed on the hub according to periodic frequencies. In this way, the blade tones of varying frequencies can be generated, and the difficulty and complexity of blade production are not increased because of the periodic frequencies.

Claims

1. A centrifugal heat dissipation fan, comprising: a housing; and an impeller, disposed in the housing, the impeller having a hub and a plurality of blades disposed surrounding the hub, wherein every two adjacent ones of the blades have air guiding inclined edges that are different from each other such that the air guiding inclined edges pass by a fixed position of the housing and generate blade tones of varying frequencies when the impeller rotates, wherein each of the air guiding inclined edges approaches the housing along with a radial direction.
2. The centrifugal heat dissipation fan according to claim 1, wherein sizes of the air guiding inclined edges show a periodic variation in accordance with a sequence in which the blades surround the hub.
3. The centrifugal heat dissipation fan according to claim 2, wherein the periodic variation comprises a sine wave function period.
4. The centrifugal heat dissipation fan according to claim 1, wherein the fixed position is near an air inlet of the housing.
5. The centrifugal heat dissipation fan according to claim 1, wherein each of the blades further has a rib located on a blade surface of the blade.
6. The centrifugal heat dissipation fan according to claim 5, wherein the rib keeps a protruding distance from the blade surface, and the protruding distance shows a periodic variation in accordance with a sequence in which the blades surround the hub.
7. The centrifugal heat dissipation fan according to claim 1, wherein the blades respectively has a combining section and an air suction section, the combining section is combined with the hub, the air suction section has the air guiding inclined edge near the combining section, and the air guiding inclined edge corresponds to an air inlet of the housing.
8. The centrifugal heat dissipation fan according to claim 7, wherein the impeller has a rotation axis, and a size of the combining section along the rotation axis is smaller than a size of the air suction section along the rotation axis.
9. The centrifugal heat dissipation fan according to claim 1, wherein the impeller has a rotation axis, and the air guiding inclined edge is inclined relative to the rotation axis and forms an included angle.
10. The centrifugal heat dissipation fan according to claim 9, wherein the included angles of the blades show a periodic variation in accordance with a sequence in which the blades surround the hub.
11. The centrifugal heat dissipation fan according to claim 7, wherein each of the blades further has an extension section between the air guiding inclined edge and the combining section, and the extension section corresponds to the air inlet.
12. The centrifugal heat dissipation fan according to claim 1, wherein the air guiding inclined edges and the housing form a tapered space, and the tapered space radially tapers from an air inlet of the housing relative to a rotation axis of the impeller.

13. The centrifugal heat dissipation fan according to claim 1, wherein the air guiding inclined edge is existed on an upper blade edge of the blade, a lower blade edge of the blade, or both of the upper and the lower blade edges.

14. The centrifugal heat dissipation fan according to claim 1, wherein the number of blades is a prime number.
